

8 (**short**) Assuming that the birth rate remains the same throughout the year and that people are born independently of each other, show that

$$P_n = \frac{d!}{(d-n)!d^n}$$

is the probability that n people chosen at random will all have different birthdays. The number of days in a year is d and n is less than d . [10]

9 (**short**) The function $f(x,y)$ is given by

$$f(x,y) = 2x^3 + 6xy^2 - 3y^3 - 150x$$

Find and classify the stationary points of $f(x,y)$. [10]

10 (**long**) An even function $f(t)$ is periodic with a period $T = 2$ and $f(t) = \cosh(t-1)$ for $0 \leq t \leq 1$.

(a) Sketch $f(t)$ in the range $-2 \leq t \leq 4$. [6]

(b) Show that

$$f(t) = \sinh(1) + 2 \sinh(1) \sum_{n=1}^{\infty} \frac{\cos n\pi t}{(1+n^2\pi^2)}$$

is a Fourier series representation of $f(t)$. [15]

(c) Using the solution in (b), deduce that

$$\sum_{n=1}^{\infty} \frac{1}{1+n^2\pi^2} = \frac{1}{e^2-1}$$

[9]